

PATIENT INFORMATION

Patient Name:	Mr. Rahul Verma	Patient ID:	APL-PT-20260091
Date of Birth:	18-April-1967 (Age: 59 years)	Gender:	Male
Blood Group:	O+	Contact:	+91-98112-67834
Address:	14, Jayanagar 4th Block, Bengaluru – 560041, Karnataka		
Referred By:	Dr. Kavitha Rao, MD (Cardiology)	Department:	Cardiology / Internal Medicine
Sample Collected:	22-May-2026, 07:15 AM	Report Generated:	22-May-2026, 12:30 PM

CLINICAL SUMMARY

Mr. Rahul Verma, a 59-year-old male, presented with complaints of exertional chest discomfort, increasing fatigue, mild ankle oedema, and occasional palpitations over the past 4–6 weeks. He has a known history of **Type 2 Diabetes Mellitus** (diagnosed 2010), **Systemic Hypertension** (diagnosed 2008), and **Dyslipidaemia** (on statin therapy since 2018). He is a reformed smoker (stopped 8 years ago; 15 pack-year history). Occasional alcohol use. Current medications: Metformin 1000mg BD, Glimepiride 2mg OD, Amlodipine 10mg OD, Losartan 50mg OD, Atorvastatin 40mg OD, Aspirin 75mg OD.

VITAL SIGNS

Blood Pressure	Heart Rate	SpO2	Temperature	Resp Rate	BMI
148/92 mmHg ↑ Elevated	82 bpm Normal	96% Normal	98.4°F Normal	17/min Normal	28.6 kg/m² ↑ Overweight

Weight: 84 kg Height: 171 cm Waist Circumference: 98 cm (↑ Elevated)

LABORATORY INVESTIGATIONS

Complete Blood Count (CBC) & ESR

Test	Result	Unit	Reference Range
Haemoglobin (Hb)	11.8 ↓	g/dL	13.5 – 17.5
RBC Count	4.0 ↓	mill/μL	4.5 – 5.5
WBC (Total Leucocyte Count)	8,200	cells/μL	4,000 – 11,000
Platelets	1,85,000	/μL	1,50,000 – 4,00,000
MCV	76.2 ↓	fL	80 – 100
MCH	25.4 ↓	pg	27 – 32
MCHC	31.6 ↓	g/dL	32 – 36
Haematocrit (PCV)	37.4 ↓	%	40 – 52
Neutrophils	66	%	50 – 70
Lymphocytes	24	%	20 – 40
Eosinophils	5	%	1 – 6

Monocytes	5	%	2 – 8
ESR	38 ↑	mm/hr	0 – 20

Biochemistry Panel

Test	Result	Unit	Reference Range
Fasting Blood Glucose	158 ↑	mg/dL	70 – 100
Post-Prandial Glucose (2hr)	224 ↑	mg/dL	< 140
HbA1c	8.4 ↑	%	< 5.7 (Normal)
Serum Creatinine	1.4 ↑	mg/dL	0.7 – 1.2
Blood Urea Nitrogen (BUN)	24 ↑	mg/dL	7 – 20
eGFR (CKD-EPI)	52 ↓	mL/min/1.73m2	> 60
Urine Albumin-Creatinine Ratio	52 ↑	mg/g	< 30
Serum Sodium (Na+)	136	mEq/L	136 – 145
Serum Potassium (K+)	4.8	mEq/L	3.5 – 5.1
Serum Uric Acid	7.8 ↑	mg/dL	3.5 – 7.2 (M)
Total Cholesterol	236 ↑	mg/dL	< 200
LDL Cholesterol	148 ↑	mg/dL	< 70 (High-risk)
HDL Cholesterol	34 ↓	mg/dL	> 40 (Male)
Triglycerides	218 ↑	mg/dL	< 150
VLDL Cholesterol	44 ↑	mg/dL	< 30
Non-HDL Cholesterol	202 ↑	mg/dL	< 130 (High-risk)
Serum Ferritin	9.6 ↓	ng/mL	12 – 300
Serum Iron	52 ↓	µg/dL	60 – 170
TIBC	408 ↑	µg/dL	250 – 370

Cardiac Risk Markers

Test	Result	Unit	Reference Range
Troponin I (High Sensitivity)	0.028	ng/mL	< 0.040
NT-proBNP	620 ↑	pg/mL	< 125 (age < 75)
CRP (High Sensitivity)	4.8 ↑	mg/L	< 3.0
Creatine Kinase (CK-MB)	28 ↑	U/L	< 25
Homocysteine	18.4 ↑	µmol/L	5 – 15
Lipoprotein (a) – Lp(a)	62 ↑	mg/dL	< 30

Thyroid Function Tests

Test	Result	Unit	Reference Range
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TSH (Thyroid Stimulating Hormone)	5.8 ↑	mIU/L	0.4 – 4.5
Free T4 (fT4)	0.88	ng/dL	0.8 – 1.8
Free T3 (fT3)	2.8	pg/mL	2.3 – 4.2

CLINICAL IMPRESSION & DIAGNOSIS

1. Microcytic Anaemia – likely Iron Deficiency (IDA)

Hb 11.8 g/dL with low MCV/MCH/MCHC, reduced ferritin and serum iron, elevated TIBC. Likely aggravated by chronic low-grade renal insufficiency (CKD Stage G3a).

2. Type 2 Diabetes Mellitus – Poorly Controlled

FBG 158 mg/dL, PPBG 224 mg/dL, HbA1c 8.4% — significantly above target (< 7.0%). Diabetic nephropathy risk supported by elevated urine albumin-creatinine ratio (52 mg/g).

3. Hypertensive Heart Disease / Elevated Cardiovascular Risk

BP 148/92 mmHg despite dual antihypertensive therapy. NT-proBNP elevated at 620 pg/mL suggesting early cardiac stress. Combined with Lp(a) 62 mg/dL, hsCRP 4.8, and homocysteine 18.4 µmol/L, 10-year ASCVD risk is HIGH. Intensified management required.

4. Dyslipidaemia – Undertreated (LDL above goal)

LDL 148 mg/dL (target < 70 mg/dL in very-high-risk patient), low HDL 34 mg/dL, elevated TG 218 mg/dL and Lp(a). Statin intensity must be escalated.

5. Chronic Kidney Disease (CKD) – Stage G3a

eGFR 52 mL/min/1.73m2 with microalbuminuria — consistent with CKD Stage G3a, likely diabetic/hypertensive nephropathy.

6. Subclinical Hypothyroidism

TSH mildly elevated at 5.8 mIU/L with normal fT4/fT3. Monitor; consider levothyroxine if symptomatic or TSH > 10.

7. Hyperuricaemia

Uric acid 7.8 mg/dL — associated with cardiovascular and renal risk. Dietary modification and monitoring advised.

TREATMENT RECOMMENDATIONS

Medications:

- Escalate statin to Rosuvastatin 40mg OD (high-intensity); add Ezetimibe 10mg if LDL target unmet at 6 weeks.
- Optimise diabetes: add Empagliflozin 10mg OD (cardio-renal protective) and refer to Endocrinology.
- Antihypertensive: add Spironolactone 25mg OD or increase Losartan to 100mg; target BP < 130/80 mmHg.
- Iron supplementation: Ferrous Sulphate 200mg OD with Vitamin C; recheck CBC and ferritin at 6 weeks.
- Subclinical hypothyroidism: monitor TSH at 3 months; initiate Levothyroxine 25mcg if TSH > 10 or symptomatic.
- Allopurinol 100mg OD for hyperuricaemia if lifestyle measures fail.
- Continue Aspirin 75mg OD for secondary cardiovascular prevention.

Cardiology Workup:

- 2D Echocardiogram to assess LV function, wall motion abnormalities, and diastolic dysfunction.
- Treadmill test (TMT) / Stress Echocardiogram for evaluation of exertional chest discomfort.
- 24-hour Holter monitoring if palpitations persist.
- Ankle-Brachial Index (ABI) screening for peripheral arterial disease.

Diet & Lifestyle:

- Diabetic, low-sodium (< 2g/day), low-purine, and heart-healthy diet.
- Mediterranean-pattern diet with increased omega-3 intake (oily fish, walnuts, flaxseed).
- Moderate aerobic exercise: 30 min walking, 5 days/week (avoid high-intensity until cardiac workup complete).

- Target weight reduction of 5–7% body weight over 6 months.
- Absolute alcohol abstinence; strict salt restriction.

Follow-up & Monitoring:

- Review in 6 weeks: CBC, FBG, HbA1c, lipid profile, serum creatinine, urine ACR, TSH.
- Nephrology referral for CKD G3a management and prevention of progression.
- Endocrinology referral for diabetes optimisation.
- Annual eye examination (diabetic retinopathy screening).
- Annual foot examination and monofilament sensory testing.

PHYSICIAN SIGN-OFF

Dr. Kavitha Rao, DM (Cardiology)
MBBS, MD, DM – Cardiology
Reg. No: KMC-2006-11204
Senior Consultant Cardiologist
Apollo Multispeciality Hospital, Bengaluru

Report Verified By:
Dr. Suresh Iyer, MD (Pathology)
MBBS, MD – Pathology & Lab Medicine
Reg. No: KMC-2003-07788
Head, Department of Clinical Pathology

This report has been digitally authenticated by the treating physician and verified by the pathology department. Results should be interpreted in clinical context. For queries, contact lab@apollohospital.in.